

PRE-REGISTRATION P7

Field-Specified Developmental Arrest of Neural Crest Migration in *Sus scrofa*: The Cross-Fostering Experiment

Domestication Coherence Gradient Field Applied from Birth
to F1 Offspring of Wild-Genome *Sus scrofa*
Predicts Morphological Shift Toward Domestic Phenotype
Without Genetic Modification

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THE PRE-REGISTERED CLAIM

The domestication syndrome is not produced by genetic modification of the organism. It is produced by the domestication coherence gradient field (N_{domestic}) arresting neural crest cell migration at the juvenile-proximate developmental stage. The genome does not change. The field changes. The developmental trajectory is arrested. The phenotype reflects that arrest.

This pre-registration specifies: the experimental design that tests this claim, the predicted outcome, the scoring method, the control conditions, and the success criteria — all locked before the experiment is run.

Primary prediction (P7): F1 offspring of wild-caught *Sus scrofa* (confirmed wild genome, never domesticated lineage) raised from birth under full N_{domestic} via domestic foster sow will show, at 12 months, significantly smaller adrenal glands and reduced HPA axis reactivity compared to genotype-matched F1 controls raised under N_{wild} . Adrenal Mass Ratio (AMR) ≤ 0.80 relative to wild-raised controls. Cortisol Reactivity Ratio (CRR) ≤ 0.75 relative to wild-raised controls. Both criteria confirmed at 12 months.

1. Background and Theoretical Basis

1.1. The Standard Model and Its Failure

The standard explanation for the domestication syndrome — the convergent suite of morphological and behavioural traits (floppy ears, depigmentation, shortened snout, smaller adrenal glands, docility, juvenile behaviour retention) appearing in every domesticated species — invokes genetic change: selective breeding shifts allele frequencies toward domestic-type variants, which alter developmental programs, which produce the domestic phenotype.

This explanation is incomplete for reasons that are quantitative and structural, not merely theoretical:

1. Marcström & Essen-Möller (1968) documented domestication syndrome traits in the first captive generation of wild boar *without selective breeding*, at frequencies inconsistent with recessive allele expression in a wild founding population.
2. The Belyaev silver fox experiment (1979) demonstrated that selecting only for HPA axis suppressibility (tameness) produces the full morphological syndrome including floppy ears, piebald coat, and shorter snout — without selecting for any morphological trait.
3. Feralization (domestic $\rightarrow$ wild) occurs in months to one generation. Re-domestication (wild $\rightarrow$ domestic) requires many generations of sustained selection. This asymmetry is not predicted by genetic models.

4. All syndrome traits map to a single developmental perturbation: mild reduction in neural crest cell (NCC) migration (Wilkins, Wrangham & Fitch, 2014). The genetic model cannot explain why every independent domestication event produces reduction in the same developmental program.

1.2. The Attractor Geometry Explanation

The attractor geometry framework proposes:

$$S + N + G \rightarrow R$$

The genome (S) encodes build capacity for both wild-type and domestic-type phenotypes. The coherence gradient field (N) specifies which developmental attractor basin the genome resolves to within the landscape geometry (G). The phenotype (R) is the output of that resolution.

The domestication coherence gradient field (N_{domestic}) acts on the maternal HPA axis, suppressing cortisol, altering the intrauterine coherence gradient environment, and arresting neural crest migration at the juvenile-proximate developmental stage. The domestic phenotype is the signature of that arrest. The genome does not change. The field changes. The developmental trajectory is arrested.

The complete theoretical derivation, literature confirmation against five independent sources, and the full experimental prediction set (P7–P11) are given in the linked theory paper: [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935).

This pre-registration tests the primary prediction of this mechanism: that applying N_{domestic} from birth to F1 offspring of a fully wild genome will shift morphology toward the domestic phenotype — specifically in the most field-responsive tissue (adrenal gland, neural crest-derived, HPA axis endpoint) — without any genetic modification.

2. Experimental Design

2.1. Overview

A cross-fostering experiment in which F1 offspring of wild-caught *Sus scrofa* are placed at birth with domestic foster sows under full N_{domestic} conditions, and compared at 3, 6, and 12 months to genotype-matched F1 controls raised under N_{wild} conditions.

All animals are genotyped at birth to confirm wild genome identity and to match genotypes across conditions. The experiment tests whether the coherence gradient field applied from birth — not genotype — determines the degree of neural crest developmental arrest.

2.2. Animal Subjects

Source animals: Wild-caught *Sus scrofa* (European wild boar or feral *Sus scrofa* with confirmed multi-generational wild ancestry, minimum 3 generations without domestic management). Source population must be confirmed by microsatellite or SNP genotyping to have no recent admixture with domestic breeds (admixture < 5%, confirmed by STRUCTURE or ADMIXTURE analysis).

F1 generation: Offspring of wild-caught sows, sired by wild-caught boars. Born and immediately assigned to experimental conditions (see Section 2.3). Minimum $n = 8$ per condition per litter cohort. Target total: $n \geq 20$ per condition across ≥ 3 independent litter cohorts.

Domestic foster sows: Commercial breed domestic sows (Large White or Landrace) with confirmed domestic breed registration. Foster sows must have successfully raised at least one previous litter and be confirmed healthy and lactating at time of fostering.

2.3. Experimental Conditions

2.3.1. Condition A: N_{domestic} from Birth (Treatment)

F1 wild-genome piglets placed with domestic foster sow within 6 hours of birth, before any wild maternal imprinting.

Field components of N_{domestic} :

- Ad libitum commercial pig feed from weaning (21 days)
- Indoor housing: temperature-regulated (18–22 °C), predator-free

- Domestic conspecific social environment (foster sow + domestic litter-mates)
- Human contact: handler interaction minimum 30 minutes twice daily from day 1
- No aversive stimuli. No predation exposure.
- Standard veterinary care throughout

2.3.2. Condition B: N_{wild} from Birth (Control)

F1 wild-genome piglets remain with wild-caught biological mother in outdoor or semi-wild enclosure.

Field components of N_{wild} :

- Natural foraging supplemented to minimum nutritional standard (welfare requirement) but not ad libitum
- Outdoor housing with shelter available: ambient temperature, no climate control
- Wild conspecific social environment (biological mother + wild litter-mates)
- Minimal human contact (necessary husbandry only)
- Environmental complexity including substrate rooting, varied terrain

2.3.3. Condition C: Domestic-Genome Control

Domestic breed piglets (Large White or Landrace) raised under identical N_{domestic} conditions as Condition A. Provides the domestic-genome domestic-field reference. Confirms that measurements in Condition A that approach Condition C values represent genuine field-specified arrest rather than measurement artefact. Target $n \geq 10$.

2.4. Genotyping Protocol

All animals genotyped within 48 hours of birth. Minimum panel: 15 microsatellite loci standard for *Sus scrofa* population genetics, plus genome-wide SNP panel (GeneSeek Genomic Profiler Porcine or equivalent) to confirm:

1. Wild ancestry (< 5% domestic admixture) for Condition A and B animals
2. Genotype matching across conditions (Conditions A and B animals selected to have equivalent allele frequency distributions at all measured loci)
3. No genetic outliers included in primary analysis

Genotyping laboratory must be blinded to experimental condition assignment.

3. Primary Outcome Measures

3.1. Primary Endpoints

The primary endpoints are neural crest-derived tissue measurements, selected because they are the most direct downstream indicators of NCC migration completion and the most rapidly responsive to field change:

3.1.1. Adrenal Gland Mass (Primary)

Measurement: Total bilateral adrenal gland mass at necropsy (12 months), normalised to body weight.

$$\text{AMR} = \frac{\bar{A}_{\text{Cond.A}}}{\bar{A}_{\text{Cond.B}}}$$

where $\bar{A}$ is mean normalised bilateral adrenal mass within condition.

Rationale: The adrenal medulla is NCC-derived (chromaffin cells originate from NCC). Adrenal medulla mass reflects the degree of NCC migration completion to the adrenal anlage during embryogenesis. This is the tissue identified by Hemmer (1990) as the earliest and most consistent indicator of domestication-state change in captive wild animals.

Protocol: Animals humanely euthanised at 12 months. Both adrenal glands dissected, trimmed of perirenal fat, weighed to nearest 0.001 g. Medulla and cortex dissected and weighed separately post-fixation (4% PFA, 48h). Medulla mass recorded as primary measurement. Body weight recorded at euthanasia. Normalised adrenal medulla mass = medulla mass / body weight × 1000 (mg/kg).

3.1.2. HPA Axis Reactivity (Primary)

Measurement: Serum cortisol response to standardised stressor at 3, 6, and 12 months.

Cortisol Reactivity Ratio:

$$\text{CRR} = \frac{\Delta[\text{cortisol}]_{\text{Cond.A}}}{\Delta[\text{cortisol}]_{\text{Cond.B}}}$$

where $\Delta[\text{cortisol}]$ is peak cortisol minus baseline cortisol at T+30 minutes post-stressor.

Stressor protocol (standardised): Animal moved to unfamiliar pen (3m×3m, novel environment, no conspecifics visible). Blood draw at T=0 (baseline, before stressor), T=30 min (peak stress response), T=60 min (recovery). Serum cortisol by validated ELISA (Enzo Life Sciences porcine cortisol kit or equivalent with intra-assay CV < 10%).

Same stressor protocol applied identically to all conditions at identical ages. Handlers blinded to condition assignment during blood draws.

3.2. Secondary Endpoints

Secondary endpoints are measured at 3, 6, and 12 months and reported as exploratory. They are not included in the primary success/failure criteria but are reported as confirmatory evidence of the mechanism.

3.2.1. Coat Pigmentation Pattern

Measurement: Percentage of total body surface area showing depigmentation (white or pale patches) at 6 and 12 months. Photographed under standardised lighting (dorsal, lateral left, lateral right views). Image analysis by blinded scorer using ImageJ colour thresholding.

$$\text{Pigmentation Score} = \frac{\text{White patch area (cm}^2\text{)}}{\text{Total body surface area (cm}^2\text{)}} \times 100$$

3.2.2. Ear Cartilage Position

Measurement: Ear position angle from vertical plane at 6 and 12 months. Photographed from anterior aspect, standardised restraint. Angle measured from ear base to ear tip relative to vertical (0° = fully erect; 90° = fully lateral). Mean of left and right ear.

3.2.3. Snout Morphology Index

Measurement: Snout length (nasal tip to orbital midpoint) divided by snout width (widest point at base of snout) at 6 and 12 months. Measured from standardised lateral and dorsal photographs using ImageJ. Higher ratio = more elongated (wild-type). Lower ratio = more abbreviated (domestic-type).

3.2.4. Hair Coarseness Score

Measurement: Tactile scoring by blinded handler (1–5 scale: 1 = fine domestic coat, 5 = coarse wild-type bristle) at dorsal mid-back location. Minimum 3 independent scorers. Mean score recorded. Inter-rater reliability confirmed (Cronbach's $\alpha > 0.80$ required before scoring begins).

3.2.5. Body Composition

Measurement: Ultrasound backfat depth at P2 position (6.5 cm lateral to dorsal midline at last rib) at 6 and 12 months. B-mode ultrasound. Higher backfat = more domestic metabolic program.

4. The Two Models and Their Diverging Predictions

This experiment is designed to distinguish two models that make identical predictions for behaviour but diverging predictions for morphology:

Outcome	Genetic model	Field-arrest model
Adrenal mass in Cond. A vs. Cond. B (identical genomes)	No difference	Smaller in Cond. A (AMR ≤ 0.80)
HPA reactivity in Cond. A vs. Cond. B	No difference	Reduced in Cond. A (CRR ≤ 0.75)
Coat depigmentation in Cond. A vs. Cond. B	No difference	Higher in Cond. A
Ear position in Cond. A vs. Cond. B	No difference	Greater droop in Cond. A
Behaviour (tameness) in Cond. A vs. Cond. B	More tame in Cond. A (learning/imprinting)	More tame in Cond. A (basin occupation)

The models agree on behaviour. Behaviour change with domestic field application from birth is explained by learning and imprinting under the genetic model and by basin occupation under the field-arrest model.

The models diverge on morphology. The genetic model predicts no morphological difference between genotype-matched animals in different fields. The field-arrest model predicts specific morphological shifts in NCC-derived tissues in the direction of the applied field.

The primary endpoints (adrenal mass, HPA reactivity) are the measurements that distinguish the models. Behaviour is not the distinguishing test. Morphology is.

5. Success Criteria

5.1. Primary Success Criterion

P7-SUCCESS requires ALL of the following at the 12-month timepoint:

Criterion 1 (Adrenal): Mean normalised adrenal medulla mass in Condition A $\leq 80\%$ of mean normalised adrenal medulla mass in Condition B. (AMR ≤ 0.80 , $p < 0.05$, two-tailed Mann-Whitney U or independent samples t -test depending on normality.)

Criterion 2 (HPA): Mean cortisol stress response (Δ cortisol, T+30 minus baseline) in Condition A $\leq 75\%$ of mean cortisol stress response in Condition B. (CRR ≤ 0.75 , $p < 0.05$.)

Criterion 3 (Genotype confirmation): No significant difference in allele frequency distribution between Condition A and Condition B animals at any locus in the genotyping panel. (Confirmed by STRUCTURE analysis prior to primary endpoint analysis.)

5.2. Partial Confirmation Criteria

If Criterion 1 is met but not Criterion 2, or vice versa, the result is recorded as **partial confirmation** with the following interpretation:

Criterion 1 met, Criterion 2 not met: Structural NCC arrest occurs (adrenal morphology shifted) but the HPA axis set point requires longer field exposure or earlier (intrauterine) field application to shift. This directs to P8 (maternal field window experiment) as the necessary follow-on test.

Criterion 2 met, Criterion 1 not met: HPA axis reactivity shifted (functional endpoint) without structural adrenal change. Suggests functional reprogramming precedes structural morphology change. Consistent with the developmental plasticity sequence prediction: functional endpoints respond faster than structural ones. Both partial outcomes are informative. Neither is a falsification of the mechanism.

6. Control Conditions and Confound Management

6.1. The Critical Controls

C1 — Genotype matching (primary confound control): The genetic model predicts no morphological difference between genotype-matched animals in different fields. Genotype matching is therefore not merely good experimental practice — it is the specific instrument that rules out the genetic model if the primary prediction confirms. Genotype confirmation must precede any primary endpoint analysis. If

genotype matching fails, the experiment must be repeated with a larger cohort before any conclusion is drawn.

C2 — Domestic-genome reference (Cond. C): Provides the domestic-field domestic-genome reference point. If Condition A approaches Condition C values, this confirms that the field is producing the same morphological outcome regardless of genome origin. If Condition A does not approach Condition C values, the field effect exists but is incomplete for postnatal-only field application — which directs to P8.

C3 — Litter cohort replication: Minimum 3 independent litter cohorts required before primary analysis is performed. Litter of origin is included as a random effect in all mixed-effects models. This controls for maternal effects within wild-type litters that are not attributable to field assignment.

C4 — Handler blinding: All morphological measurements, blood draws, and sample processing must be performed by handlers blinded to condition assignment. Genotyping laboratory must be blinded. Image analysis must be blinded. Blinding is documented and confirmed before any analysis.

C5 — Nutrition equivalence: Condition A and B animals must receive nutritionally equivalent diets from the point of weaning (21 days). Differences in ad libitum provision are a field component (resource security is part of N_{domestic}) but differences in total nutritional intake could confound body composition endpoints. Total caloric intake must be monitored and reported. Body weight at each timepoint is a covariate in all models.

6.2. Pre-specified Confound Interpretations

Confound: Litter size differences. Wild litters are typically larger (6–12 piglets) than domestic litters selected for fostering (4–8 piglets from domestic sow). Per-piglet nutritional access may differ. Control: weigh animals at every timepoint; include body weight as covariate.

Confound: Foster rejection. Domestic sow may reject wild-genome piglets due to odour differences. Protocol: introduce piglets within 6 hours of birth during foster sow's own parturition or within 48 hours of a litter she has accepted. Document all rejection events. Replace foster sow if rejection persists > 24 hours. Rejection rate reported in results.

Confound: Epigenetic carryover from wild mother. The F1 animals' epigenome was established in utero under N_{wild} (wild mother). This is not a confound — it is the experimental condition. The prediction is that postnatal N_{domestic} will shift morphology despite the intrauterine wild field. If the effect is absent, it directs to P8 (intrauterine field required). If the effect is present, it demonstrates that postnatal field is sufficient for at least partial arrest. The intrauterine starting condition is identical across Condition A and B animals and is not a confound for the between-condition comparison.

7. Failure Diagnosis

If P7 fails (no morphological difference between Condition A and Condition B with confirmed genotype matching), the following diagnoses are pre-specified:

Diagnosis F1 (Critical window missed): The NCC migration critical window is prenatal, not postnatal. Postnatal field application cannot arrest an already-completed developmental program. *Direction:* Proceed immediately to P8 (maternal field from conception experiment). P7 failure under this diagnosis does not falsify the field-arrest model. It specifies the critical window more precisely.

Diagnosis F2 (Field specification incomplete): The domestic field as applied (indoor housing, ad libitum food, human contact) is not the complete field specification required to produce the arrest. A missing field component (e.g., specific microbiome, specific tactile stimulation, specific olfactory environment from domestic conspecifics from conception) is required. *Direction:* Compare field component specification between Condition A and known domestication events. Identify the missing component. Redesign with complete field.

Diagnosis F3 (Genetic floor effect): The wild-genome animals have a deeper Basin A than the field-arrest model's current specification predicts. The field applied is sufficient to arrest a less deep Basin A (as in heritage domestic breeds) but not sufficient to arrest the wild-caught *Sus scrofa* Basin A within one postnatal generation. *Direction:* Repeat with heritage breed wild-type animals (Mangalitsa, Ossabaw

Island) which have shallower Basin A due to prior partial domestication selection. A gradient of effect across wild-genome depth is a prediction of the model (see P10).

Diagnosis F4 (Genuine falsification): If F1, F2, and F3 are all ruled out by appropriate follow-on experiments, and no morphological shift occurs under any condition of field specification, the field-arrest model is falsified for the domestication syndrome. The genetic model is the correct explanation. This outcome would be published in full.

8. Scoring Schedule and Timeline

Timepoint	Measurements
Birth (T=0)	Genotyping (blood). Body weight. Foster assignment documented. Photograph (dorsal, lateral bilateral).
3 months	Body weight. Hair coarseness score (blinded). Ear position angle (photograph). Cortisol stress response (blood draw protocol). Coat pigmentation photograph.
6 months	Body weight. Hair coarseness score. Ear position angle. Snout morphology index. Cortisol stress response. Coat pigmentation. Backfat ultrasound (P2).
12 months	Body weight. All secondary endpoints repeated. Cortisol stress response (primary endpoint). Necropsy: bilateral adrenal glands dissected, weighed, fixed (4% PFA). Adrenal medulla/cortex separated and weighed post-fixation (primary endpoint). Histology: H&E on adrenal sections. IHC: chromogranin A (medulla marker), SF-1 (cortex marker).

Minimum timeline from litter birth to primary endpoint: 12 months.

Minimum cohort timeline including 3-cohort replication: 18–24 months (cohorts staggered, not sequential).

Estimated cost per cohort: Animal acquisition and housing (12 months): \$8,000–\$15,000 per condition. Genotyping panel: \$2,000–\$4,000. ELISA consumables: \$1,500–\$3,000. Histology and IHC: \$3,000–\$6,000. Total per cohort: \$25,000–\$45,000 (3 conditions). Total for 3-cohort replication: \$75,000–\$135,000.

9. Statistical Analysis Plan

All analyses are pre-specified. No unplanned analyses will be reported as confirmatory. Exploratory analyses will be clearly labelled.

Primary analysis (adrenal mass, HPA reactivity): Linear mixed-effects model with condition (A, B, C) as fixed effect, litter cohort as random effect, body weight at 12 months as covariate. Primary comparison: Condition A vs. Condition B.

Normality: Shapiro-Wilk test on residuals. If non-normal: Mann-Whitney U test for primary comparison; Kruskal-Wallis for three-condition comparison.

Effect size: Cohen’s *d* (parametric) or rank-biserial correlation *r* (non-parametric) reported for all primary comparisons.

Primary significance threshold: $\alpha = 0.05$, two-tailed. Bonferroni correction applied for multiple primary endpoints (AMR and CRR): adjusted $\alpha = 0.025$ per endpoint.

Secondary endpoints: All secondary endpoints analysed identically but reported as exploratory. No success/failure determination made from secondary endpoints alone.

Longitudinal analysis: Repeated measures for 3-month, 6-month, and 12-month cortisol data. Mixed-effects model with time as a within-subject factor. Interaction term (condition $\times$ time) reported to assess whether field effect increases over the measurement period.

Pre-analysis lock: This statistical analysis plan is locked at pre-registration. Any deviation from it in the final publication must be declared in the methods section with justification.

10. Ethical Compliance

This experiment involves vertebrate animals. The following ethical requirements apply:

- Institutional Animal Care and Use Committee (IACUC) or equivalent national ethics body approval required before any animal work begins.
- All animals must meet or exceed the minimum welfare standards of the applicable national legislation throughout the experiment.
- Condition B animals (wild-type field) must receive supplemental nutrition sufficient to maintain body weight within 10% of Condition A animals. The field distinction is in security, not starvation.
- Euthanasia at 12 months by trained personnel using approved methods (captive bolt or barbiturate overdose per institutional protocol).
- Foster rejection events require immediate supplemental feeding within 2 hours and veterinary assessment. Rejected piglets must not be left without nutrition for > 4 hours.
- Wild-caught source animals must be legally obtained with appropriate wildlife licences for the jurisdiction of collection.

No genetic modification is performed on any animal in this experiment. The experiment uses only naturally occurring genomes and environmental field manipulation.

11. Linked Records

This pre-registration is P7 in the OrganismCore experimental series. The following records existed at the time of this deposit and are cited here with confirmed DOIs. No forward references to undeclared documents are included.

ID	Title	DOI
Theory paper	Domestication as Field-Specified Developmental Arrest	10.5281/zenodo.19094935
P-INVERSION-1	Plant Architecture Inversion via Coherence Gradient Field Specification	10.5281/zenodo.19040399
LECA-A	LECA Resurrection Fungal Arm	10.5281/zenodo.18986790
FOXA1/EZH2	FOXA1/EZH2 Ratio as Universal Breast Cancer Subtype Classifier	10.5281/zenodo.18883922

12. Reporting Standard

Results of this experiment will be reported against this pre-registration regardless of outcome. Specifically:

1. If P7 succeeds (both criteria met): The report will state that the primary prediction was confirmed, that the genetic model’s null prediction was rejected, and that the field-arrest mechanism is supported.
2. If P7 partially confirms (one criterion met): The report will state which criterion was met, apply the pre-specified partial interpretation, and proceed to the directed follow-on experiment.
3. If P7 fails (neither criterion met): The report will apply the failure diagnosis protocol (Section 7), state which diagnosis applies, and proceed to the directed follow-on experiment. The failure will be published in full. A failure in which F1, F2, and F3 are subsequently also ruled out constitutes a falsification of the field-arrest model, which will be published as such.

Selective reporting — publishing only the confirming result if multiple outcome measures were taken — is explicitly prohibited. All pre-specified endpoints are reported in all publications arising from this pre-registration.

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Repository: [github.com/Eric-Robert-Lawson/ attractor-oncology](https://github.com/Eric-Robert-Lawson/attractor-oncology)

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